

Fig. 7 supplement | Deployment decision flow

Audit outcomes translated into conservative validation and deployment actions.

Audit pattern	Decision category	Recommended action
Raw high + background-centered high + acceptable calibration	 candidate for controlled deployment	Proceed only with local calibration check, locked thresholds, and ongoing drift monitoring.
Raw high + background-centered high + poor calibration	 ranking only recalibrate before clinical use	Use only as a ranking signal until calibration, thresholds, and clinical operating points are locally reset.
Raw high + background-centered low	 background dependent retrain locally	Do not deploy as portable focal-drug model; validate locally and retrain or recalibrate on current ecology.
Raw high + underpowered matching	 insufficient matched evidence	Collect more matched support or additional external labels before making a deployment claim.
Raw low + background-centered low	 not deployment ready	Do not deploy; use for research triage or redesign the model/evaluation target.